

ECE 4375: Automatic Control Systems

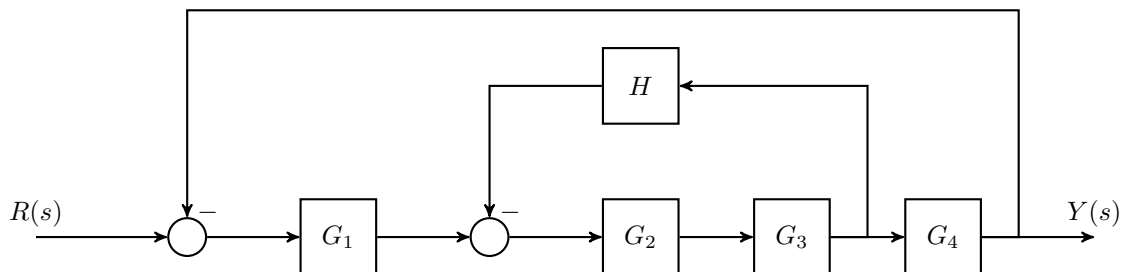
Homework Assignment 2

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Date: 9/16/26

Instructions: Show all work for full credit. Box your final numerical or symbolic answers where applicable. Unless otherwise stated, assume all initial conditions are zero.

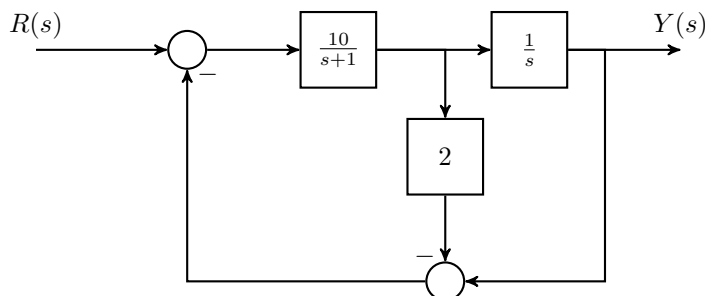
1. Given:



Find:

- (a) The general open-loop transfer function, $\frac{Y(s)}{R(s)} = \frac{B(s)}{A(s)}$
- (b) Set $G_1 = \frac{1}{s}$, $G_2 = \frac{1}{s+1}$, $G_3 = \frac{1}{s}$, $G_4 = \frac{2}{s+2}$, and $H = \frac{5}{s+5}$ then determine the open-loop transfer function, $\frac{Y(s)}{R(s)} = \frac{B(s)}{A(s)}$

2. Given:



Find:

- (a) Open-loop transfer function, $\frac{Y(s)}{R(s)} = \frac{B(s)}{A(s)}$
- (b) Closed-loop transfer functions G and H where, $\frac{Y(s)}{R(s)} = \frac{G(s)}{1+GH(s)}$

3. Given:

$$\frac{Y(s)}{U(s)} = \frac{4(s+50)}{s^2+30s+200}$$

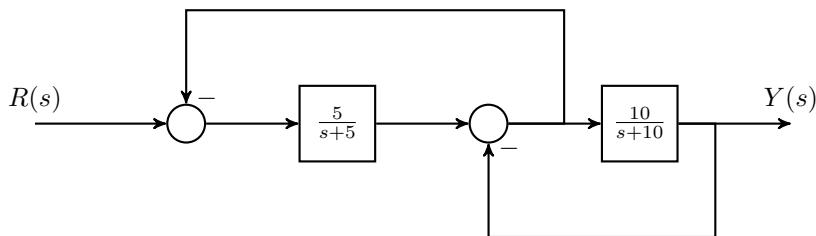
- (a) Find $y(\infty)$ when the input is:
- $u(t)$ is an impulse
 - $u(t)$ is a unit step
- (b) Determine a block diagram of the equation

4. Given:

$$\frac{Y(s)}{U(s)} = \frac{s^2-4s+6}{s^2+3s+6}$$

- (a) Find $y(\infty)$ when the input is:
- $u(t)$ is an impulse
 - $u(t)$ is a unit step
- (b) Find a block diagram of the equation

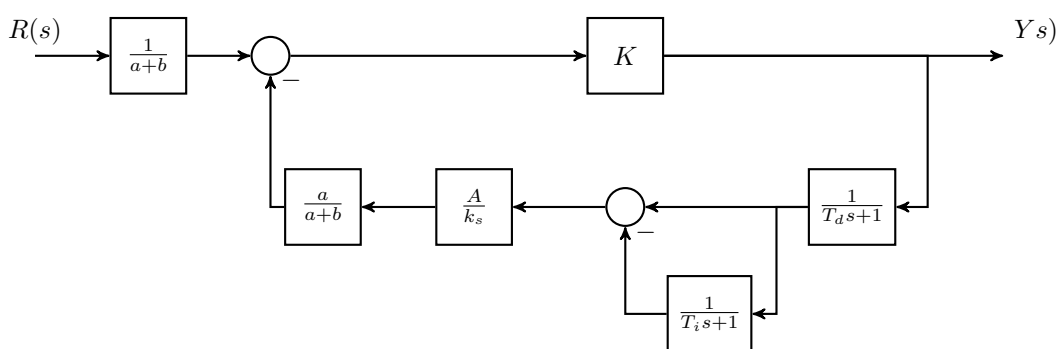
5. Given:



Find:

- (a) The transfer function $\frac{Y(s)}{R(s)}$
- (b) Closed-loop transfer functions G and H where, $\frac{Y(s)}{R(s)} = \frac{G(s)}{1+GH(s)}$

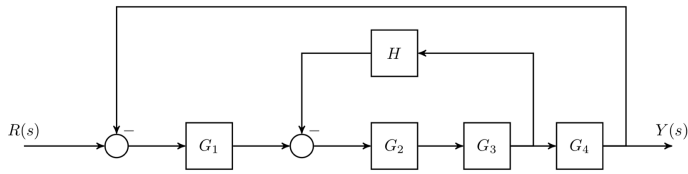
6. Given:



Find:

- (a) The transfer function, $\frac{Y(s)}{R(s)}$
- (b) Set: $a = 1$, $b = 3$, $T_i = 10$, $T_d = 1$, $A = 2$, $k_s = 4$, and $K = 1$ then find the poles and zeros of the system.

1. Given:



Find:

- (a) The general open-loop transfer function, $\frac{Y(s)}{R(s)} = \frac{B(s)}{A(s)}$
 (b) Set $G_1 = \frac{1}{s}$, $G_2 = \frac{1}{s+1}$, $G_3 = \frac{1}{s}$, $G_4 = \frac{2}{s+2}$, and $H = \frac{5}{s+5}$ then determine the open-loop transfer function, $\frac{Y(s)}{R(s)} = \frac{B(s)}{A(s)}$

a)

$$F(s) = G_1 \cdot \frac{G_2 G_3}{1 + G_2 G_3 H} \cdot G_4$$

$$R(s) \rightarrow G_1 \rightarrow \boxed{\frac{G_2 G_3}{1 + G_2 G_3 H}} \rightarrow G_4 \rightarrow Y(s)$$

$$\frac{Y(s)}{R(s)} = \frac{G_1 G_2 G_3 G_4}{1 + G_2 G_3 H + G_1 G_2 G_3 G_4}$$

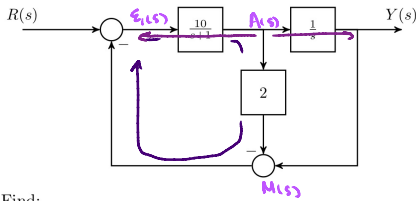
b) $G_1 = \frac{1}{s}$, $G_2 = \frac{1}{s+1}$, $G_3 = \frac{1}{s}$, $G_4 = \frac{2}{s+2}$, $H = \frac{5}{s+5}$

$$G_1 G_2 G_3 G_4 = \frac{2}{s^2(s+1)(s+2)} \quad , \quad G_2 G_3 H = \frac{5}{s(s+1)(s+5)}$$

$$D = s^2(s+1)(s+2)(s+5) = s^5 + 8s^4 + 17s^3 + 15s^2 + 12s + 10$$

$$\frac{Y(s)}{R(s)} = \frac{2s + 10}{s^5 + 8s^4 + 17s^3 + 15s^2 + 12s + 10}$$

2. Given:



Find:

- (a) Open-loop transfer function, $\frac{Y(s)}{R(s)} = \frac{B(s)}{A(s)}$
 (b) Closed-loop transfer functions G and H where, $\frac{Y(s)}{R(s)} = \frac{G(s)}{1 + GH(s)}$

a) Open Loop - $\frac{Y(s)}{R(s)} = \frac{B(s)}{A(s)}$

b) $\frac{Y(s)}{R(s)} = \frac{G(s)}{1 + GH(s)}$
 $\frac{1}{T(s)} = \frac{s^2 - 19s + 10}{10} = \frac{s^2 - 19s}{10} + \frac{1}{10}$

$$E_1(s) = R(s) - M(s)$$

$$M(s) = Y(s) - 2A(s)$$

$$A(s) = \frac{10}{s+1} E_1(s) = A(s) = sY(s)$$

$$\frac{1}{s} A(s)$$

$$G(s) = \frac{s^2 - 19s}{1}$$

$$H(s) = 1$$

$$E_1(s) = R(s) - M(s) = R(s) - Y(s) + 2A(s)$$

$$A(s) = \frac{10}{s+1} [R(s) - Y(s) + 2A(s)]$$

$$sY(s) = \frac{10}{s+1} [R(s) - Y(s) + 2sY(s)]$$

$$sY(s)(s+1) = 10R(s) - 10Y(s) + 20sY(s)$$

$$s^2 Y(s) + sY(s) + 10Y(s) - 20sY(s) = 10R(s)$$

$$s^2 Y(s) + sY(s) + 10Y(s) - 20sY(s) = 10R(s)$$

$$s^2 Y(s) + sY(s) + 10Y(s) - 20sY(s) = 10R(s)$$

$$\frac{Y(s)(s^2 - 19s + 10)}{R(s)} = \frac{10R(s)}{R(s)} = \frac{10}{s^2 - 19s + 10}$$

3. Given:

$$\frac{Y(s)}{U(s)} = \frac{4(s+50)}{s^2+30s+200}$$

(a) Find $y(\infty)$ when the input is:

i. $u(t)$ is an impulse

ii. $u(t)$ is a unit step

(b) Determine a block diagram of the equation

$$\frac{Y(s)}{U(s)} = \frac{4(s+50)}{s^2+30s+200}$$

poles: $s^2+30s+200=0$
 $(s+10)(s+20) \rightarrow$
 $s=-10 \quad s=-20$

a i) $U(s)=1$
 $u(t) \rightarrow \lim_{s \rightarrow \infty} s \cdot \frac{4(s+50)}{s^2+30s+200} \cdot 1 \rightarrow 0 \cdot \frac{200}{200} = 0$

a ii) Unit step input: $U(s)=\frac{1}{s}$

$$y(\infty) = \lim_{s \rightarrow 0} s \cdot \frac{4(s+50)}{s^2+30s+200} \cdot \frac{1}{s} = \frac{4(50)}{200} = 1$$

b) Block diagram

$$U(s) \longrightarrow \boxed{\frac{4(s+50)}{s^2+30s+200}} \longrightarrow Y(s)$$

4. Given:

$$\frac{Y(s)}{U(s)} = \frac{s^2-4s+6}{s^2+3s+6}$$

$$s^2+3s+6 \rightarrow \text{poles = complex}$$

real = $-\frac{3}{2}$

(a) Find $y(\infty)$ when the input is:

i. $u(t)$ is an impulse ✓

ii. $u(t)$ is a unit step ✓

(b) Find a block diagram of the equation

b)

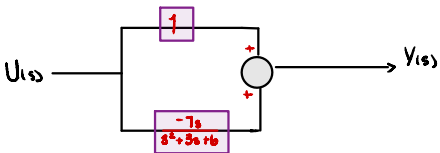
$$T(s) = \frac{Y(s)}{U(s)} = \frac{s^2-4s+6}{s^2+3s+6}$$

$$T(s) = 1 + \frac{-7s}{s^2+3s+6}$$

instant feed through

$\frac{-7s}{s^2+3s+6}$ dyn. remain.

$$Y(s) = 1 \cdot U(s) + \frac{-7s}{s^2+3s+6} \cdot U(s)$$



$$T(s) = \frac{Y(s)}{U(s)} = \frac{s^2-4s+6}{s^2+3s+6}$$

quadratic = $\frac{b \pm \sqrt{b^2-4ac}}{2}$

$$s^2+3s+6=0 \rightarrow \frac{-3 \pm \sqrt{9-24}}{2} = \frac{-3 \pm \sqrt{-15}}{2} = \frac{-3 \pm j\sqrt{15}}{2}$$

real part

a) i) $u(t)$ impulse

$$U(s)=1$$

$$Y(s) = T(s) \cdot 1 = \frac{s^2-4s+6}{s^2+3s+6}$$

$$y(\infty) = \lim_{s \rightarrow 0} s Y(s) = \lim_{s \rightarrow 0} s \cdot \frac{s^2-4s+6}{s^2+3s+6}$$

$s=0$

$$T(0) = \frac{0-0+6}{0+0+6} = \frac{6}{6} = 1$$

$$y(\infty) = \lim_{s \rightarrow 0} s \cdot 1 = 0$$

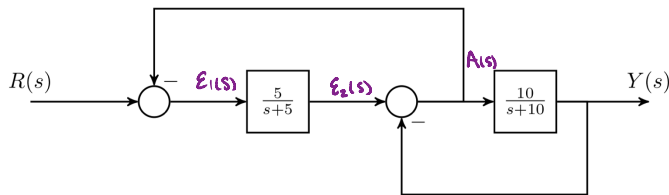
ii) $U(s)=\frac{1}{s}$

$$Y(s) = T(s) \cdot \frac{1}{s} = \frac{s^2-4s+6}{s(s^2+3s+6)}$$

$$y(\infty) = \lim_{s \rightarrow 0} s \cdot Y(s) = \lim_{s \rightarrow 0} s \cdot \frac{s^2-4s+6}{s(s^2+3s+6)}$$

$$y(\infty) = \lim_{s \rightarrow 0} \frac{s^2-4s+6}{s^2+3s+6} = T(0) = \frac{6}{6} = 1$$

5. Given:



Find:

(a) The transfer function $\frac{Y(s)}{R(s)}$

(b) Closed-loop transfer functions G and H where $\frac{Y(s)}{R(s)} = \frac{G(s)}{1+GH(s)}$

$$b) \frac{1}{T(s)} = \frac{s^2 + 30s + 150}{50} \rightarrow \frac{s^2 + 30s}{50} + \frac{150}{50}$$

$G(s) \quad H(s)$

$$T(s) = \frac{1}{\frac{1}{G(s)} + H(s)} = \frac{50}{s^2 + 30s} + 3$$

$$= \frac{50}{s(s+30)} + 3$$

$$\downarrow$$

$$\frac{G}{1+GH(s)} = \frac{\frac{50}{s(s+30)}}{1 + \frac{50}{s(s+30)} \cdot 3} = \frac{50}{s(s+30) + 150} = \frac{50}{s^2 + 30s + 150}$$

a)

$$E_1 = R(s) - A(s) \leftarrow \text{this is sum 1 after the outer feedback tap A}$$

$$A(s) = E_2(s) - Y(s) \leftarrow \text{this is sum 2 the inner feedback tap Y}$$

$$E_2(s) = \frac{5}{s+5} E_1(s) \quad Y(s) = \frac{10}{s+10} A(s)$$

$$A = \frac{s+10}{10} Y$$

$$E_1 = R - \frac{s+10}{10} Y$$

$$E_2 = \frac{5}{s+5} \left[R - \frac{s+10}{10} Y \right]$$

$$(s+5) \frac{s+10}{10} Y = \frac{5}{s+5} \left[R - \frac{s+10}{10} Y \right] - Y$$

$$\frac{(s+5)(s+10)}{10} Y = 5R - \frac{5(s+10)}{10} Y - (s+5) Y$$

$$\frac{(s+10)(s+5)}{10} Y + \frac{s+10}{2} Y + (s+5) Y = 5R \cdot 10$$

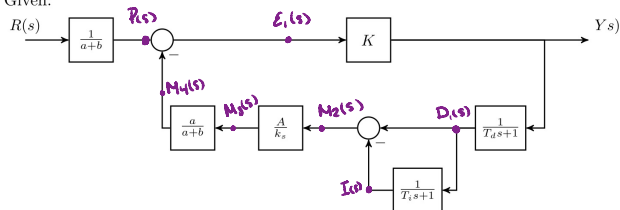
$$Y[(s+10)(s+5) + 5(s+10) + 10(s+5)] = 50R$$

$$s^2 + 15s + 50 + 5s + 50 + 10s + 50 = 50R$$

$$Y[s^2 + 30s + 150] = 50R$$

$$\frac{Y(s)}{R(s)} = \frac{50}{s^2 + 30s + 150}$$

6. Given:



$$E_1(s) \rightarrow K \rightarrow Y(s)$$

$$Y(s) = \frac{1}{T_d s + 1} \rightarrow D(s)$$

$$D = \frac{Y}{T_d s + 1}$$

$$M_2 = D \cdot \frac{T_i s}{T_i s + 1}$$

Find:

(a) The transfer function, $\frac{Y(s)}{R(s)}$

(b) Set: $a = 1$, $b = 3$, $T_i = 10$, $T_d = 1$, $A = 2$, $k_s = 4$, and $K = 1$ then find the poles and zeros of the system.

$$\frac{Y(s)}{R(s)} = \frac{K k_s (T_d s + 1)(T_i s + 1)}{(a+b)k_s(T_d s + 1)(T_i s + 1) + K a A T_i s}$$

$\leftarrow \text{forward}$ $\leftarrow \text{feedback}$ $\leftarrow \text{rest of the loop}$

$$T(s) = \frac{40s^2 + 44s + 4}{100s^2 + 196s + 16}$$

$$\frac{10s^2 + 11s + 1}{40s^2 + 44s + 4} \rightarrow (s+1)(10s+1)$$

$$s = \frac{-49 \pm \sqrt{49^2 - 4(40)(4)}}{2(40)} \rightarrow \frac{-49 \pm \sqrt{2401 - 640}}{80} \rightarrow \frac{-49 \pm 41.96}{80}$$

$$s = \frac{-49 + 41.96}{80} = \frac{-7.04}{80} = -0.088$$

$$s = \frac{-49 - 41.96}{80} = \frac{-90.96}{80} = -1.137$$

poles
 $s = -0.088, -1.137$

$$a) \frac{Y}{(T_d s + 1)(T_i s + 1)}; M_2 = \frac{Y}{T_d s + 1} - \frac{Y}{(T_d s + 1)(T_i s + 1)}$$

$$M_2 = \frac{Y}{T_d s + 1} \left[1 - \frac{1}{T_i s + 1} \right] = \frac{Y}{T_d s + 1} \cdot \frac{(T_i s + 1) - 1}{T_i s + 1}$$

$$M_2 = \frac{Y}{T_d s + 1} \cdot \frac{T_i s}{T_i s + 1} = \frac{Y \cdot T_i s}{(T_d s + 1)(T_i s + 1)}$$

$$M_3 = \frac{A}{k_s} \cdot M_2; M_4 = \frac{a}{a+b} \cdot \frac{A}{k_s} \cdot \frac{T_i s Y}{(T_d s + 1)(T_i s + 1)}$$

$$F(s) = \frac{a A T_i s}{(a+b)k_s(T_d s + 1)(T_i s + 1)} = M_4 = F(s) \cdot Y$$

$$E_1 = \frac{R}{a+b} - M_4 = \frac{R}{a+b} - F(s) Y$$

$$Y = K E_1 = K \left[\frac{R}{a+b} - F(s) Y \right] = \frac{KR}{a+b} - K F(s) Y$$

$$Y + K F(s) Y = \frac{KR}{a+b}$$

$$Y \left[1 + K F(s) \right] = \frac{KR}{a+b} = \frac{Y}{R} = \frac{K}{\frac{a+b}{KR}}$$

$$\frac{Y(s)}{R(s)} = \frac{\frac{K}{a+b}}{1 + \frac{K a A T_i s}{(a+b)k_s(T_d s + 1)(T_i s + 1)}} = \frac{K k_s (T_d s + 1)(T_i s + 1)}{(a+b)k_s(T_d s + 1)(T_i s + 1) + K a A T_i s}$$

$\leftarrow \text{forward}$ $\leftarrow \text{feedback}$ $\leftarrow \text{rest of the loop}$